



# Logic Gate Simulator

Mohammad Labak posted Oct 29, 2024 12:06 PM • 31 Words ★ Subscribed

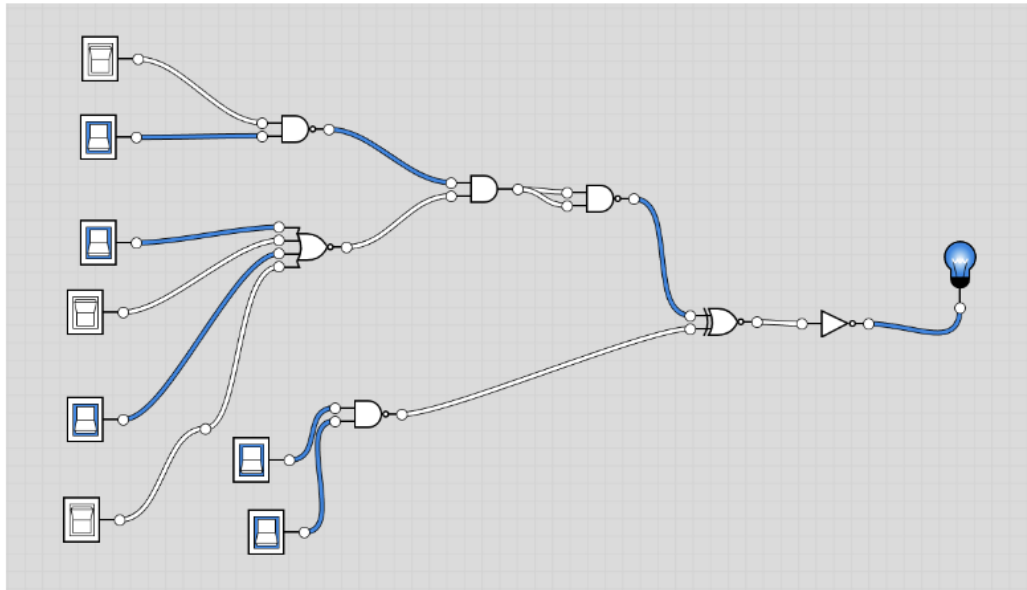


0



This website lets you manually build a digital logic circuit. Its helpful to test your circuits, to make sure they match up with your truth table.

This is the website: <https://logic.ly/demo/>



[image.png](#)

(58.56 KB)



▶ ⏮ 🔊 0:24 / 1:27



# One day, a computer will fit on a desk (1974) | RetroFocus



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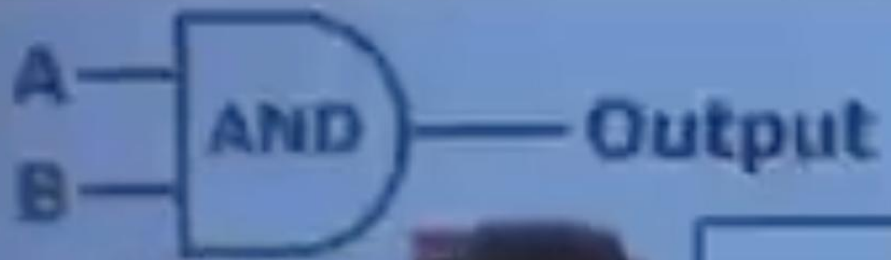


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# Logic Map

| A | B | A AND B |
|---|---|---------|
|   |   |         |
|   |   |         |
|   |   |         |
|   |   |         |
|   |   |         |
|   |   |         |
|   |   |         |

Mute (m)



exam deliver system is one way its confuse to submit the exam. i prefer that exam only on paper

A couple critics would be is try to explain things with examples, also lectures are not really good because its more about attending lectures but id recommend having a separate lecture for students to use for preparing for the midterm/ exam because the way i studied was from the textbook but it took a long time. Also maybe tips and tricks for some parts would be helpful/ more efficient ways to find the answer

Midterm was very lengthy and less time

Overall, I really appreciate the effort that is put into making sure the lectures and labs are beneficial for students' learning and not just about checking off another box towards a degree. However, I found the midterm to be quite incongruous with the philosophy of the rest of the course, as it tested mainly on the speed that students could answer relatively basic questions rather than the breadth or depth of their understanding of the material.

I understand that the time limit had to be tight enough that students could not look up all the answers, but with time being that limited and the questions being relatively straightforward, it felt like the main place I likely lost marks was based on the speed at which I typed, how many corners I wanted to risk cutting in my justifications, and how I optimized my time against the marks of each question. In other words, I felt like I was being tested more on my ability to write academic tests and play the system rather than my understanding of the course material.

I found that the midterm exam required more time to complete. Although I had prepared well and understood the material, I needed additional time to fully show my work and demonstrate my problem-solving process. Additionally, while I appreciated the open-book format, I was unsure about which online resources were permitted. For instance, there are non-AI-based conversion tools available online that function similarly to ChatGPT, but I wasn't sure if using them was allowed—so I chose not to. It would be helpful to have clearer guidelines in the future about exactly which tools, especially AI-related ones, are and aren't permitted during assessments.

Honestly, the only thing I can think of was the midterm. It wasn't necessarily hard but the combination of question 2 and question 4 made it too long. Other than that, no complaints.

To be fair, I don't really have any large critique to put forward. However, I do have a minor thing to mention about the midterm. I understand that time scarcity was a core part of the midterm's challenge, attempting to properly gauge whether we really understood and knew the underlying material. Personally, I think I would have preferred a potentially longer, traditional closed-book examination instead.

I absolutely love how you cover all your topics and go through every detail by connecting it to a real world scenario. I enjoy attending every lecture even during the summer! One thing I could say is that I understand the minimalist PowerPoints, but for the future could you at least provide some context to the slide so if we were to refer to them later for memory refresh we can understand what's happening a little more clearly. Overall, one of the best profs I've had! Thank you!

maybe your outfits😂

More time for exam please.

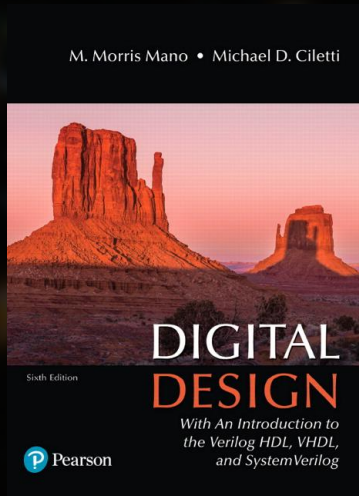
Very well!! It's just I really wish I had more time on the exam like I wasted a lot of time on question 4 and moved on from it realizing I didnt have much time left, i know for sure I didnt do the other questions properly because I was doing them in a rush and my handwriting was so bad throughout the entire exam.

I wish we had more time on the exam, I had wasted a lot of time on question 4 and started rushing through the entire rest of the exam

Everything so far so good.

you are explaining the topics clearly. Thank you !

Content is often not covered before assignments are posted and lectures are difficult to follow



# Chapter 3

## Gate-Level Minimization

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# MINIMIZATION

- I) Boolean Algebra (algebraically)  
aka. Algebraic Manipulation
-

$$F(Y_2, Y_1, X_2, X_1) = \prod M(4, 8, 9, 12, 13, 14)$$

$$= (M4)(M8)(M9)(M12)(M13)(M14)$$

$$= (Y_2 + Y'_1 + X_2 + X_1)(Y'_2 + Y_1 + X_2 + X_1)(Y'_2 + Y'_1 + X_2 + X_1)(Y'_2 + Y'_1 + X_2 + X'_1)(Y'_2 + Y'_1 + X'_2 + X_1)$$

$$\text{Dual} \rightarrow (Y_2 Y'_1 X_2 X_1) + (Y'_2 Y_1 X_2 X_1) + (Y'_2 Y'_1 X_2 X_1) + (Y'_2 Y'_1 X_2 X'_1) + (Y'_2 Y'_1 X'_2 X_1)$$

$$= (Y_2 Y'_1 X_2 X_1) + (Y'_2 Y_1 X_2 X_1) + (Y'_2 Y'_1 X_2 X_1) + (Y'_2 Y'_1 X_2 X'_1) + (Y'_2 Y'_1 X'_2 X_1)$$

$$= (Y_2 Y'_1 X_2 X_1) + (Y'_2 X_2 X_1 (Y_1 + Y'_1)) + (Y'_2 Y'_1 X_2 X'_1) + (Y'_2 Y'_1 X'_2 X_1)$$

$$= (Y_2 Y'_1 X_2 X_1) + (Y'_2 X_2 X_1 (1)) + (Y'_2 Y'_1 X_2 X'_1) + (Y'_2 Y'_1 X'_2 X_1)$$

$$= (Y_2 Y'_1 X_2 X_1) + (Y'_2 X_2 X_1) + (Y'_2 Y'_1 X_2 X'_1) + (Y'_2 Y'_1 X'_2 X_1)$$

$$= (Y_2 Y'_1 X_2 X_1) + (Y'_2 X_2 X_1) + (Y'_2 Y'_1 X_2 X'_1) + (Y'_2 Y'_1 X'_2 X_1) + (Y'_2 Y_1 X_2 X_1)$$

$$= (Y_2 Y'_1 X_2 X_1) + (Y'_2 X_2 X_1) + (Y'_2 Y'_1 X_2 X'_1) + (Y'_2 Y'_1 X'_2 X_1) + (Y'_2 Y'_1 X_2 X_1)$$

$$= (Y_2 Y'_1 X_2 X_1) + (Y'_2 X_2 X_1) + (Y'_2 Y'_1 X_2 (X'_1 + X_1)) + (Y'_2 Y'_1 X'_2 X_1)$$

$$= (Y_2 Y'_1 X_2 X_1) + (X_2 X_1 Y'_2) + (Y'_2 Y'_1 X_2 (1)) + (Y'_2 Y'_1 X'_2 X_1)$$

$$= (Y_2 Y'_1 X_2 X_1) + (X_2 X_1 Y'_2) + (Y'_2 Y'_1 X_2) + (Y'_2 Y'_1 X'_2 X_1)$$

Are we done?! Honestly, I don't know ☹️ 😊

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# MINIMIZATION

## II) Map (Karnaugh map, K-map) aka. Graphical Manipulation

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Algorithm; Straightforward, up to six variables

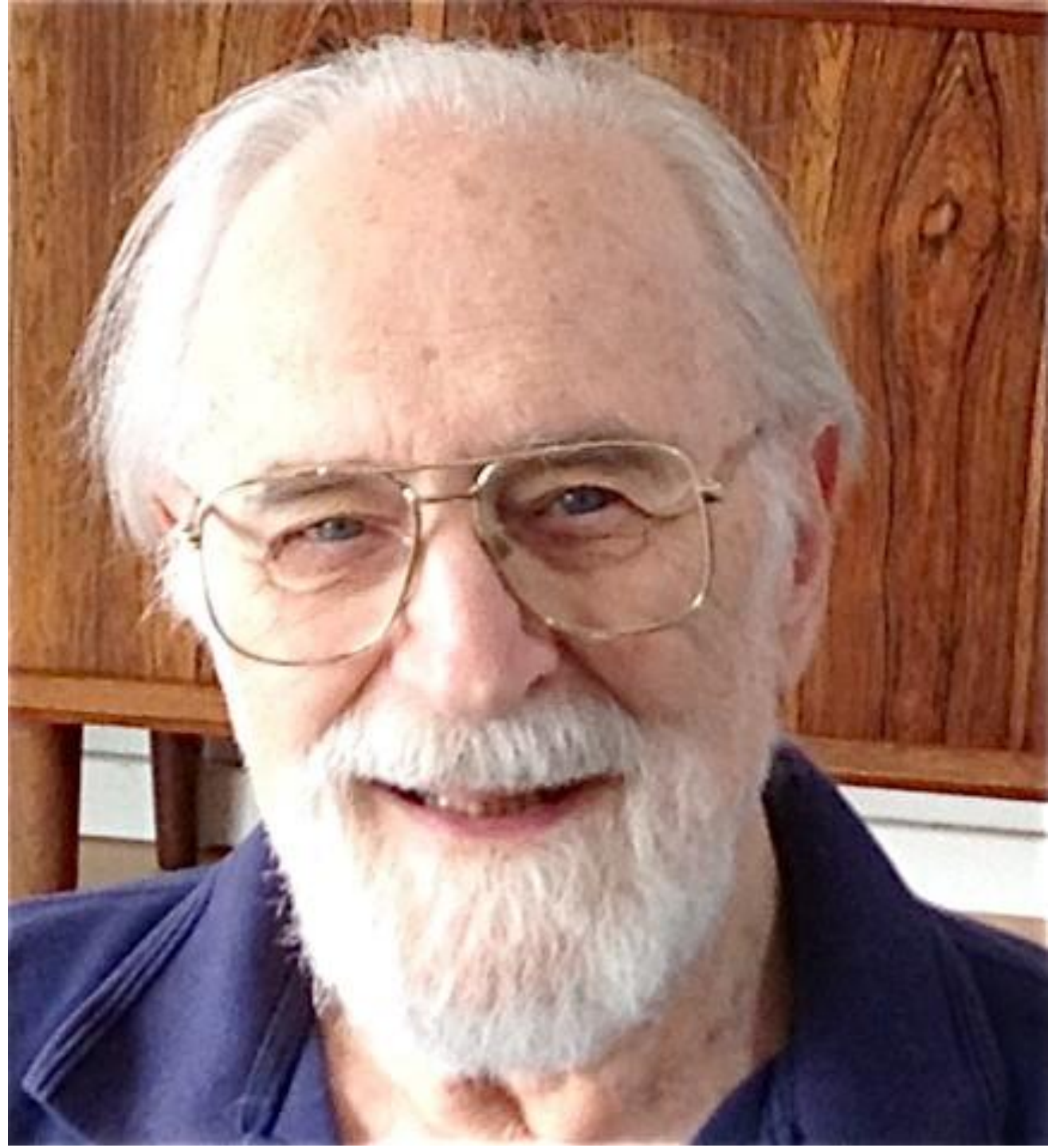
Result is always minimal

# Maurice Karnaugh

Physicist  
Mathematician  
Inventor

Bell Labs (1954)

"The Map Method for Synthesis of  
Combinational Logic Circuits"



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# 1-Variable KARNAUGH MAP

---

| X | F     |
|---|-------|
| 0 | $m_0$ |
| 1 | $m_1$ |

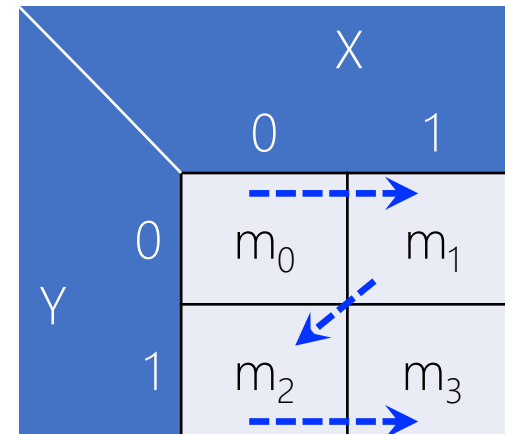
| X     |       |
|-------|-------|
| 0     | 1     |
| $m_0$ | $m_1$ |

---

# 2-Variable KARNAUGH MAP

---

| Y | X | F     |
|---|---|-------|
| 0 | 0 | $m_0$ |
| 0 | 1 | $m_1$ |
| 1 | 0 | $m_2$ |
| 1 | 1 | $m_3$ |



| Y | X | F |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

$$\begin{aligned}
 F(Y,X) &= m_0 + m_1 + m_2 \\
 &= Y'X' + Y'X + YX' \\
 &= Y'X' + Y'X' + Y'X + YX' \\
 &= Y'(X' + X) + Y'X' + YX' \\
 &= Y' + Y'X' + YX' \\
 &= Y' + X'(Y' + Y) \\
 &= Y' + X'
 \end{aligned}$$

|   |   | X |   |
|---|---|---|---|
|   |   | 0 | 1 |
| Y | 0 | 1 | 1 |
|   | 1 | 1 | 0 |

$$F(Y,X) = Y' + X'$$

| Y | X | F |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

$$\begin{aligned}
 F(Y,X) &= m_1 + m_2 \\
 &= Y'X + YX'
 \end{aligned}$$

XOR

|   |   | X |   |
|---|---|---|---|
|   |   | 0 | 1 |
| Y | 0 | 0 | 1 |
|   | 1 | 1 | 0 |

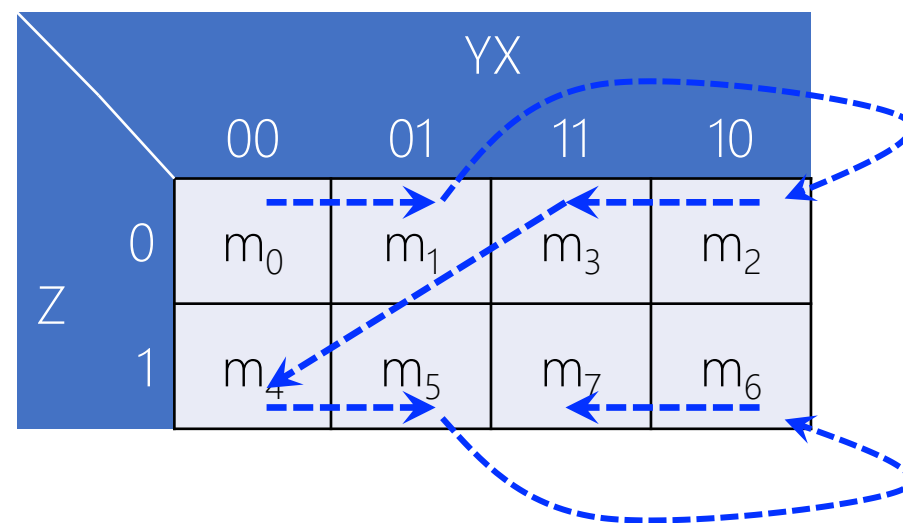
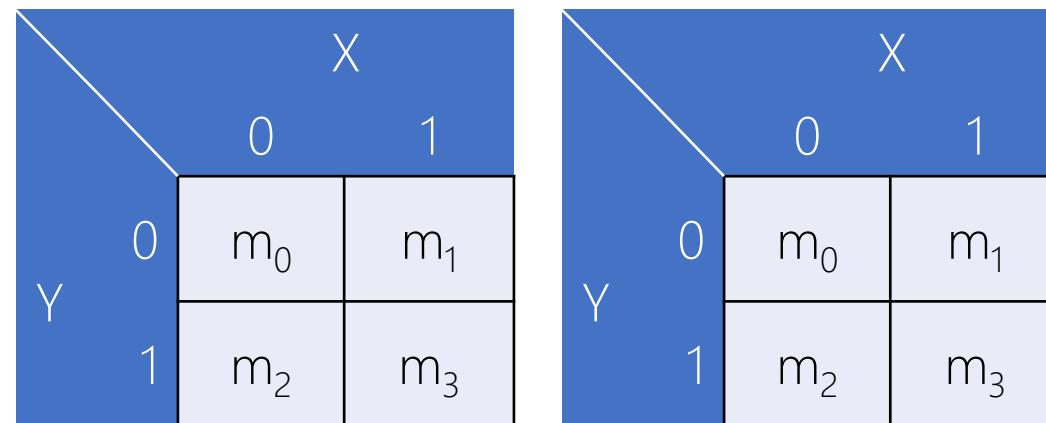
$$\begin{aligned}
 F(Y,X) &= m_1 + m_2 \\
 &= Y'X + YX'
 \end{aligned}$$

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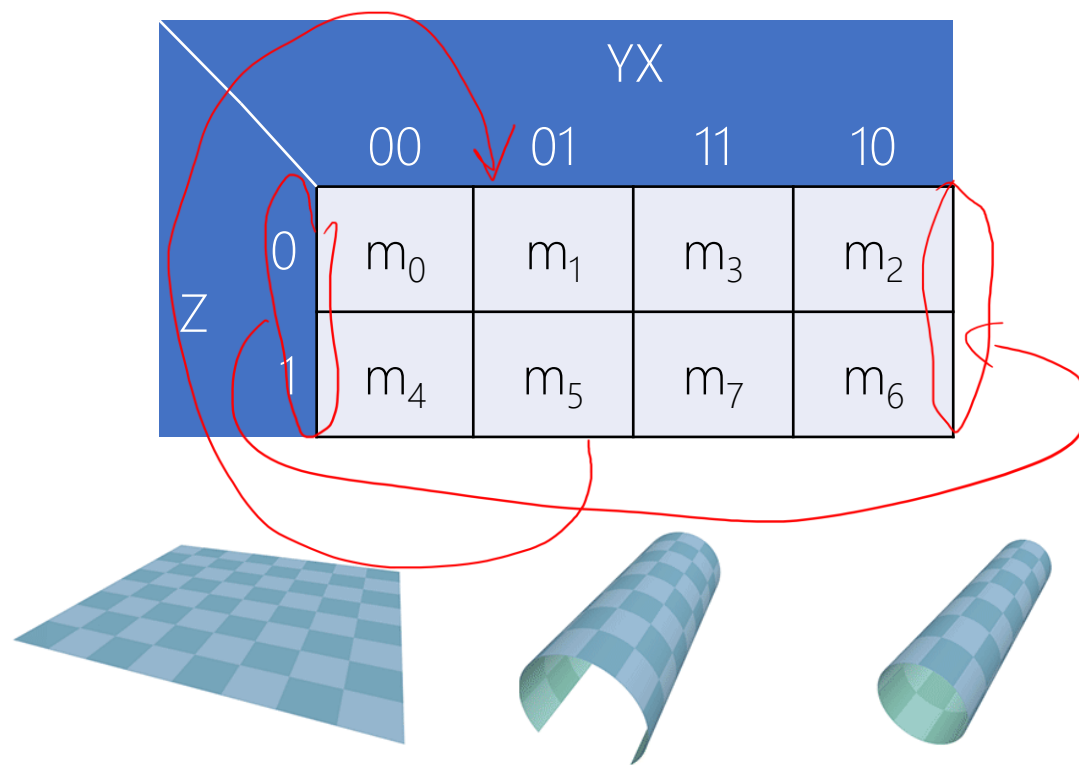
# 3-Variable KARNAUGH MAP

---

| Z | Y | X | F     |
|---|---|---|-------|
| 0 | 0 | 0 | $m_0$ |
| 0 | 0 | 1 | $m_1$ |
| 0 | 1 | 0 | $m_2$ |
| 0 | 1 | 1 | $m_3$ |
| 1 | 0 | 0 | $m_4$ |
| 1 | 0 | 1 | $m_5$ |
| 1 | 1 | 0 | $m_6$ |
| 1 | 1 | 1 | $m_7$ |



| Z | Y | X | F     |
|---|---|---|-------|
| 0 | 0 | 0 | $m_0$ |
| 0 | 0 | 1 | $m_1$ |
| 0 | 1 | 0 | $m_2$ |
| 0 | 1 | 1 | $m_3$ |
| 1 | 0 | 0 | $m_4$ |
| 1 | 0 | 1 | $m_5$ |
| 1 | 1 | 0 | $m_6$ |
| 1 | 1 | 1 | $m_7$ |



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# Click to Play!

[https://en.wikipedia.org/wiki/Karnaugh\\_map#/media/File:Torus\\_from\\_rectangle.gif](https://en.wikipedia.org/wiki/Karnaugh_map#/media/File:Torus_from_rectangle.gif)

---

| Z | Y | X | F |
|---|---|---|---|
| 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 0 |

$$\begin{aligned}
 F(Z,Y,X) &= \sum m(0,2,4, 6) \\
 &= Z'Y'X' + Z'YX' + ZY'X' + ZYX' \\
 &= ?
 \end{aligned}$$

|   |   | YX |    |    |    |
|---|---|----|----|----|----|
|   |   | 00 | 01 | 11 | 10 |
| Z | 0 | 1  | 0  | 0  | 1  |
|   | 1 | 1  | 0  | 0  | 1  |

| Z | Y | X | F |
|---|---|---|---|
| 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 0 |

$$\begin{aligned}
 F(Z,Y,X) &= \sum m(0,2,4, 6) \\
 &= Z'Y'X' + Z'YX' + ZY'X' + ZYX' \\
 &= ?
 \end{aligned}$$

|   |   | YX |    |    |    |
|---|---|----|----|----|----|
|   |   | 00 | 01 | 11 | 10 |
| Z | 0 | 1  | 0  | 0  | 1  |
|   | 1 | 1  | 0  | 0  | 1  |

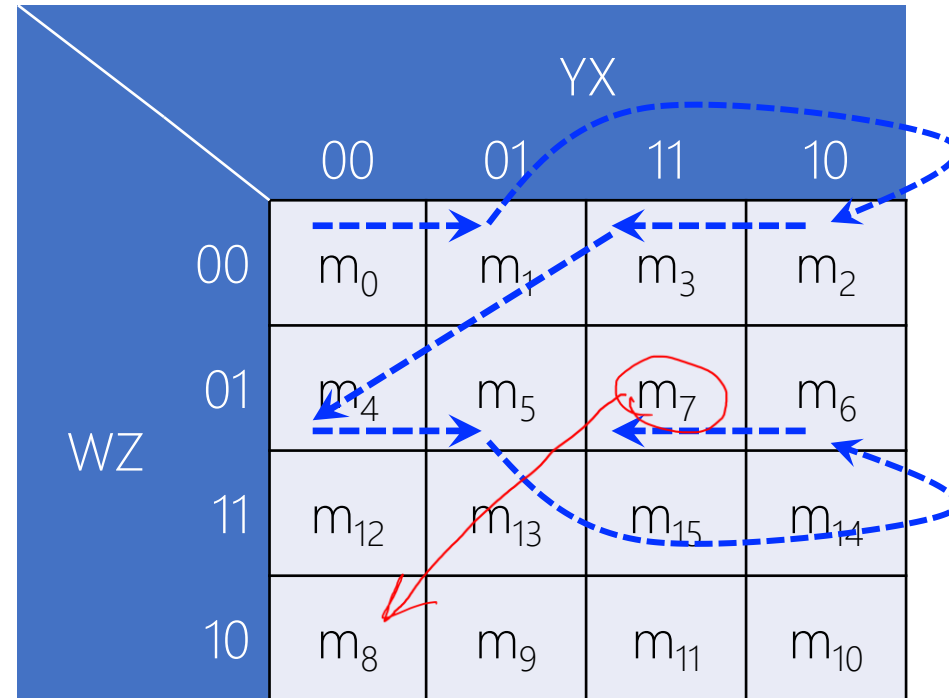
$$\begin{aligned}
 F(Z,Y,X) &= \sum m(0,2,4, 6) \\
 &= X'
 \end{aligned}$$

---

# 4-Variable KARNAUGH MAP

---

|    |    | YX       |          |          |          |
|----|----|----------|----------|----------|----------|
|    |    | 00       | 01       | 11       | 10       |
| WZ | 00 | $m_0$    | $m_1$    | $m_3$    | $m_2$    |
|    | 01 | $m_4$    | $m_5$    | $m_7$    | $m_6$    |
|    | 11 | $m_{12}$ | $m_{13}$ | $m_{15}$ | $m_{14}$ |
|    | 10 | $m_8$    | $m_9$    | $m_{11}$ | $m_{10}$ |



|    |           | YX       |          |           |          |
|----|-----------|----------|----------|-----------|----------|
|    |           | 00       | 01       | <u>11</u> | 10       |
| WZ | 00        | $m_0$    | $m_1$    | $m_3$     | $m_2$    |
|    | 01        | $m_4$    | $m_5$    | $m_7$     | $m_6$    |
|    | <u>11</u> | $m_{12}$ | $m_{13}$ | $m_{15}$  | $m_{14}$ |
|    | 10        | $m_8$    | $m_9$    | $m_{11}$  | $m_{10}$ |

|    |    | YX            |               |               |               |
|----|----|---------------|---------------|---------------|---------------|
|    |    | 00            | 01            | 11            | 10            |
| WZ | 00 | 1<br>$m_0$    | 1<br>$m_1$    | 0<br>$m_3$    | 1<br>$m_2$    |
|    | 01 | 1<br>$m_4$    | 1<br>$m_5$    | 0<br>$m_7$    | 1<br>$m_6$    |
|    | 11 | 1<br>$m_{12}$ | 1<br>$m_{13}$ | 0<br>$m_{15}$ | 1<br>$m_{14}$ |
|    | 10 | 1<br>$m_8$    | 1<br>$m_9$    | 0<br>$m_{11}$ | 0<br>$m_{10}$ |

$$F(W,Z,Y,X) = \sum m(0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14)$$

|    |    | YX                   |                      |                      |                      |
|----|----|----------------------|----------------------|----------------------|----------------------|
|    |    | 00                   | 01                   | 11                   | 10                   |
| WZ | 00 | 1<br>m <sub>0</sub>  | 1<br>m <sub>1</sub>  | 0<br>m <sub>3</sub>  | 1<br>m <sub>2</sub>  |
|    | 01 | 1<br>m <sub>4</sub>  | 1<br>m <sub>5</sub>  | 0<br>m <sub>7</sub>  | 1<br>m <sub>6</sub>  |
|    | 11 | 1<br>m <sub>12</sub> | 1<br>m <sub>13</sub> | 0<br>m <sub>15</sub> | 1<br>m <sub>14</sub> |
|    | 10 | 1<br>m <sub>8</sub>  | 1<br>m <sub>9</sub>  | 0<br>m <sub>11</sub> | 0<br>m <sub>10</sub> |

$$F(W,Z,Y,X) = \sum m(0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14)$$

$$= Y' +$$

|    |    | YX            |               |               |               |
|----|----|---------------|---------------|---------------|---------------|
|    |    | 00            | 01            | 11            | 10            |
| WZ | 00 | 1<br>$m_0$    | 1<br>$m_1$    | 0<br>$m_3$    | 1<br>$m_2$    |
|    | 01 | 1<br>$m_4$    | 1<br>$m_5$    | 0<br>$m_7$    | 1<br>$m_6$    |
|    | 11 | 1<br>$m_{12}$ | 1<br>$m_{13}$ | 0<br>$m_{15}$ | 1<br>$m_{14}$ |
|    | 10 | 1<br>$m_8$    | 1<br>$m_9$    | 0<br>$m_{11}$ | 0<br>$m_{10}$ |

Warning!

$W=\{0,1\}$ , the value of  $W$  matters in  $m_{10}$

$Z=\{0,1\}$

$Y=\{1\}$

$X=\{0\}$

$$F(W,Z,Y,X) = \sum m(0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14)$$

$$= Y' +$$

|    |    | YX            |               |               |               |
|----|----|---------------|---------------|---------------|---------------|
|    |    | 00            | 01            | 11            | 10            |
| WZ | 00 | 1<br>$m_0$    | 1<br>$m_1$    | 0<br>$m_3$    | 1<br>$m_2$    |
|    | 01 | 1<br>$m_4$    | 1<br>$m_5$    | 0<br>$m_7$    | 1<br>$m_6$    |
|    | 11 | 1<br>$m_{12}$ | 1<br>$m_{13}$ | 0<br>$m_{15}$ | 1<br>$m_{14}$ |
|    | 10 | 1<br>$m_8$    | 1<br>$m_9$    | 0<br>$m_{11}$ | 0<br>$m_{10}$ |

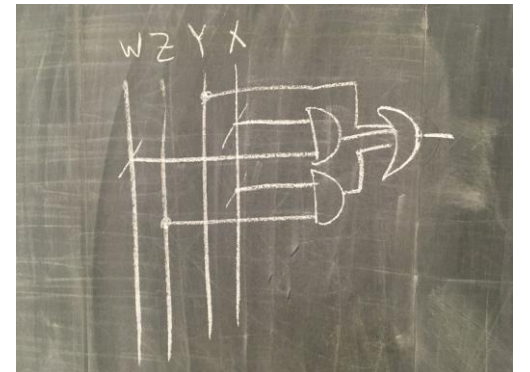
$$\begin{aligned}
 F(W,Z,Y,X) &= \sum m(0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14) \\
 &= Y' + W'YX' + ZYX'
 \end{aligned}$$

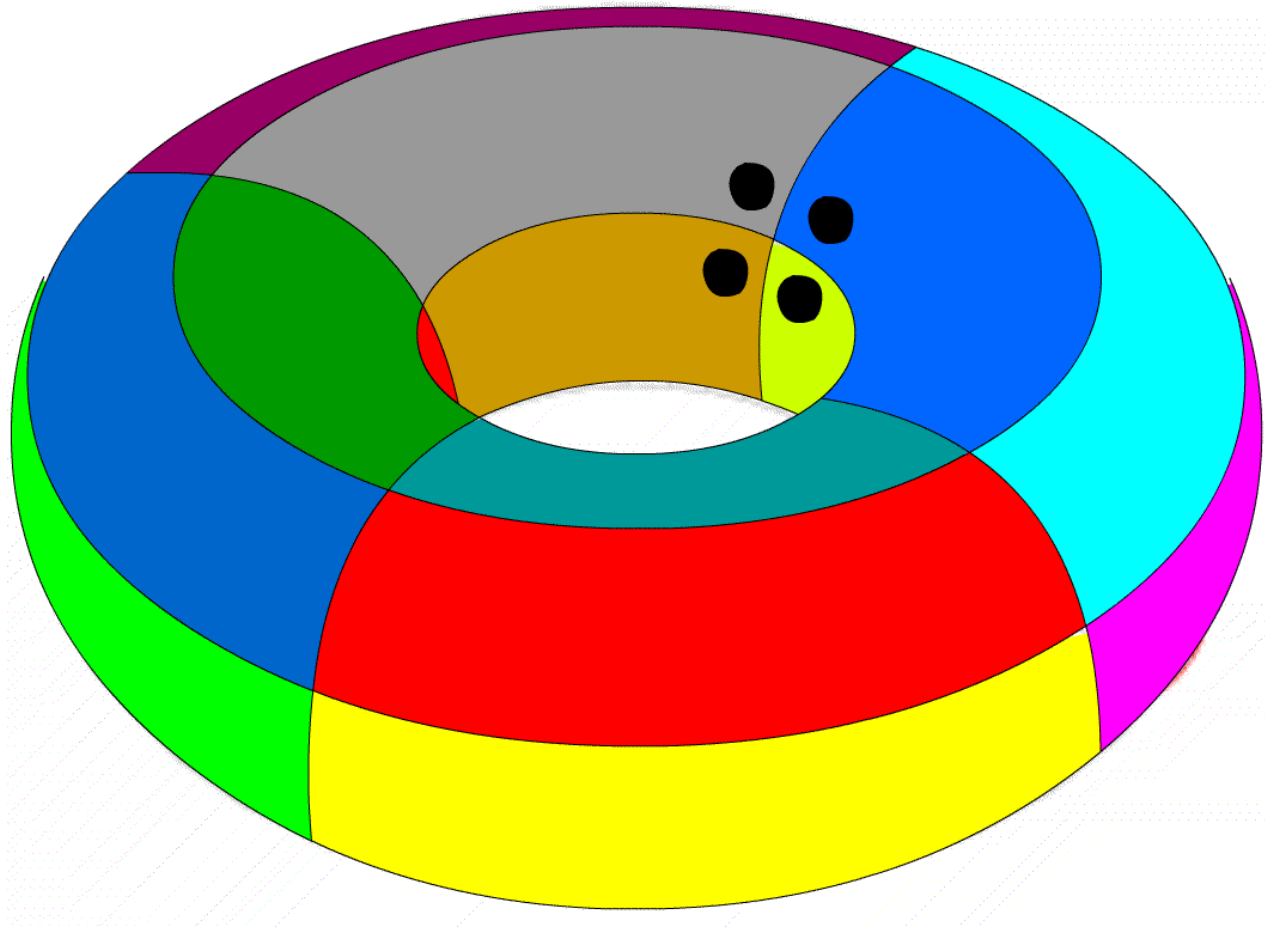
|    |    | YX            |               |               |               |
|----|----|---------------|---------------|---------------|---------------|
|    |    | 00            | 01            | 11            | 10            |
| WZ | 00 | 1<br>$m_0$    | 1<br>$m_1$    | 0<br>$m_3$    | 1<br>$m_2$    |
|    | 01 | 1<br>$m_4$    | 1<br>$m_5$    | 0<br>$m_7$    | 1<br>$m_6$    |
|    | 11 | 1<br>$m_{12}$ | 1<br>$m_{13}$ | 0<br>$m_{15}$ | 1<br>$m_{14}$ |
|    | 10 | 1<br>$m_8$    | 1<br>$m_9$    | 0<br>$m_{11}$ | 0<br>$m_{10}$ |

$$\begin{aligned}
 F(W,Z,Y,X) &= \sum m(0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14) \\
 &= Y' + W'X' + WYX'
 \end{aligned}$$

|    |    | YX            |               |               |               |
|----|----|---------------|---------------|---------------|---------------|
|    |    | 00            | 01            | 11            | 10            |
| WZ | 00 | 1<br>$m_0$    | 1<br>$m_1$    | 0<br>$m_3$    | 1<br>$m_2$    |
|    | 01 | 1<br>$m_4$    | 1<br>$m_5$    | 0<br>$m_7$    | 1<br>$m_6$    |
|    | 11 | 1<br>$m_{12}$ | 1<br>$m_{13}$ | 0<br>$m_{15}$ | 1<br>$m_{14}$ |
|    | 10 | 1<br>$m_8$    | 1<br>$m_9$    | 0<br>$m_{11}$ | 0<br>$m_{10}$ |

$$\begin{aligned}
 F(W,Z,Y,X) &= \sum m(0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14) \\
 &= Y' + W'X' + ZX'
 \end{aligned}$$





torus

|    |    | YX                   |                      |                      |                      |
|----|----|----------------------|----------------------|----------------------|----------------------|
|    |    | 00                   | 01                   | 11                   | 10                   |
| WZ | 00 | 1<br>m <sub>0</sub>  | 0<br>m <sub>1</sub>  | 0<br>m <sub>3</sub>  | 1<br>m <sub>2</sub>  |
|    | 01 | 0<br>m <sub>4</sub>  | 0<br>m <sub>5</sub>  | 0<br>m <sub>7</sub>  | 0<br>m <sub>6</sub>  |
|    | 11 | 0<br>m <sub>12</sub> | 0<br>m <sub>13</sub> | 0<br>m <sub>15</sub> | 0<br>m <sub>14</sub> |
|    | 10 | 1<br>m <sub>8</sub>  | 0<br>m <sub>9</sub>  | 0<br>m <sub>11</sub> | 1<br>m <sub>10</sub> |

$$F(W, Z, Y, X) = \sum m(0, 2, 8, 10) \\ = Z'X'$$

---

# MAXTERMS

---

| Z | Y | X | F |
|---|---|---|---|
| 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 1 |

$$\begin{aligned}
 F(Z,Y,X) &= \prod M(4,5) \\
 &= (Z' + Y + X) (Z' + Y + X') \\
 &= ?
 \end{aligned}$$

|   |   | YX |    |    |    |
|---|---|----|----|----|----|
|   |   | 00 | 01 | 11 | 10 |
| Z | 0 | 1  | 1  | 1  | 1  |
|   | 1 | 0  | 0  | 1  | 1  |

$$\begin{aligned}
 F(Z,Y,X) &= \prod M(4,5) \\
 &= ?
 \end{aligned}$$

| Z | Y | X | F |
|---|---|---|---|
| 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 1 |

$$\begin{aligned}
 F(Z,Y,X) &= \prod M(4,5) \\
 &= (Z' + Y + X) (Z' + Y + X') \\
 &= ?
 \end{aligned}$$

|   |   | YX |    |    |    |
|---|---|----|----|----|----|
|   |   | 00 | 01 | 11 | 10 |
| Z | 0 | 1  | 1  | 1  | 1  |
|   | 1 | 0  | 0  | 1  | 1  |

$$\begin{aligned}
 F(Z,Y,X) &= \prod M(4,5) \\
 &= ?
 \end{aligned}$$

| Z | Y | X | F |
|---|---|---|---|
| 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 1 |

$$\begin{aligned}
 F(Z,Y,X) &= \prod M(4,5) \\
 &= (Z' + Y + X) (Z' + Y + X') \\
 &= ?
 \end{aligned}$$

|   |   | YX |    |    |    |
|---|---|----|----|----|----|
|   |   | 00 | 01 | 11 | 10 |
| Z | 0 | 0  | 0  | 0  | 0  |
|   | 1 | 1  | 1  | 0  | 0  |

$$\begin{aligned}
 F'(Z,Y,X) &= \sum m(4,5) \\
 &= ZY'
 \end{aligned}$$

| Z | Y | X | F |
|---|---|---|---|
| 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 1 |

$$\begin{aligned}
 F(Z,Y,X) &= \prod M(4,5) \\
 &= (Z' + Y + X) (Z' + Y + X') \\
 &= ?
 \end{aligned}$$

|   |   | YX |    |    |    |
|---|---|----|----|----|----|
|   |   | 00 | 01 | 11 | 10 |
| Z | 0 | 1  | 1  | 1  | 1  |
|   | 1 | 0  | 0  | 1  | 1  |

$$\begin{aligned}
 F(Z,Y,X) &= \prod M(4,5) \\
 &= (F')' \\
 &= (ZY')' \\
 &= Z' + Y
 \end{aligned}$$

| Z | Y | X | F |
|---|---|---|---|
| 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 1 |

$$\begin{aligned}
 F(Z,Y,X) &= \sum m(0,1,2,3,6,7) \\
 &= Z'Y'X' + Z'Y'X + Z'YX' + Z'YX + ZYX' + ZYX \\
 &= ?
 \end{aligned}$$

$$\begin{aligned}
 F(Z,Y,X) &= \prod M(4,5) \\
 &= (Z' + Y + X) (Z' + Y + X') \\
 &= ?
 \end{aligned}$$

|   |   | YX |    |    |    |
|---|---|----|----|----|----|
|   |   | 00 | 01 | 11 | 10 |
| Z | 0 | 1  | 1  | 1  | 1  |
|   | 1 | 0  | 0  | 1  | 1  |

$$\begin{aligned}
 F(Z,Y,X) &= \sum m(0,1,2,3,6,7) \\
 &= Z' + Y
 \end{aligned}$$

|   |   | YX |    |    |    |
|---|---|----|----|----|----|
|   |   | 00 | 01 | 11 | 10 |
| Z | 0 | 1  | 1  | 1  | 1  |
|   | 1 | 0  | 0  | 1  | 1  |

$$\begin{aligned}
 F(Z,Y,X) &= \prod M(4,5) \\
 &= (F')' \\
 &= (ZY')' \\
 &= Z' + Y
 \end{aligned}$$

| Z | Y | X | F |
|---|---|---|---|
| 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 0 |

$$\begin{aligned}
 F(Z,Y,X) &= \prod M(1,3,5,7) \\
 &= (Z+Y+X')(Z+Y'+X')(Z'+Y+X')(Z'+Y'+X') \\
 &= ?
 \end{aligned}$$

|   |   | YX |    |    |    |
|---|---|----|----|----|----|
|   |   | 00 | 01 | 11 | 10 |
| Z | 0 | 1  | 0  | 0  | 1  |
|   | 1 | 1  | 0  | 0  | 1  |

$$\begin{aligned}
 F(Z,Y,X) &= \prod M(1,3,5,7) \\
 &= (X)' \\
 &= X'
 \end{aligned}$$

| Z | Y | X | F |
|---|---|---|---|
| 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 0 |

$$\begin{aligned}
 F(Z,Y,X) &= \prod M(1,3,5,7) \\
 &= (Z+Y+X')(Z+Y'+X')(Z'+Y+X')(Z'+Y'+X') \\
 &= ?
 \end{aligned}$$

|   |   | YX |    |    |    |
|---|---|----|----|----|----|
|   |   | 00 | 01 | 11 | 10 |
| Z | 0 | 1  | 0  | 0  | 1  |
|   | 1 | 1  | 0  | 0  | 1  |

$$\begin{aligned}
 F(Z,Y,X) &= \sum m(0,2,4,6) \\
 &= X'
 \end{aligned}$$

|   |   | YX |    |    |    |
|---|---|----|----|----|----|
|   |   | 00 | 01 | 11 | 10 |
| Z | 0 | 1  | 0  | 0  | 1  |
|   | 1 | 1  | 0  | 0  | 1  |

$$\begin{aligned}
 F(Z,Y,X) &= \prod M(1,3,5,7) \\
 &= (X)' \\
 &= X'
 \end{aligned}$$

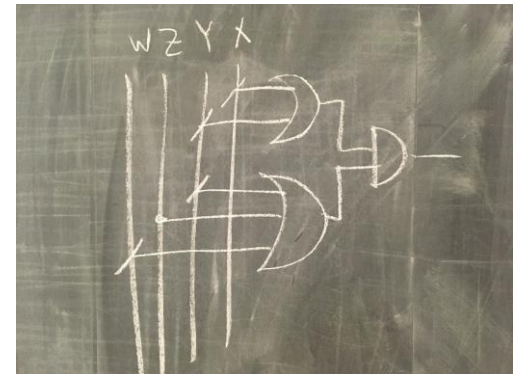
---

# MAXTERMS

---

|    |    | YX            |               |               |               |
|----|----|---------------|---------------|---------------|---------------|
|    |    | 00            | 01            | 11            | 10            |
| WZ | 00 | 1<br>$m_0$    | 1<br>$m_1$    | 0<br>$m_3$    | 1<br>$m_2$    |
|    | 01 | 1<br>$m_4$    | 1<br>$m_5$    | 0<br>$m_7$    | 1<br>$m_6$    |
|    | 11 | 1<br>$m_{12}$ | 1<br>$m_{13}$ | 0<br>$m_{15}$ | 1<br>$m_{14}$ |
|    | 10 | 1<br>$m_8$    | 1<br>$m_9$    | 0<br>$m_{11}$ | 0<br>$m_{10}$ |

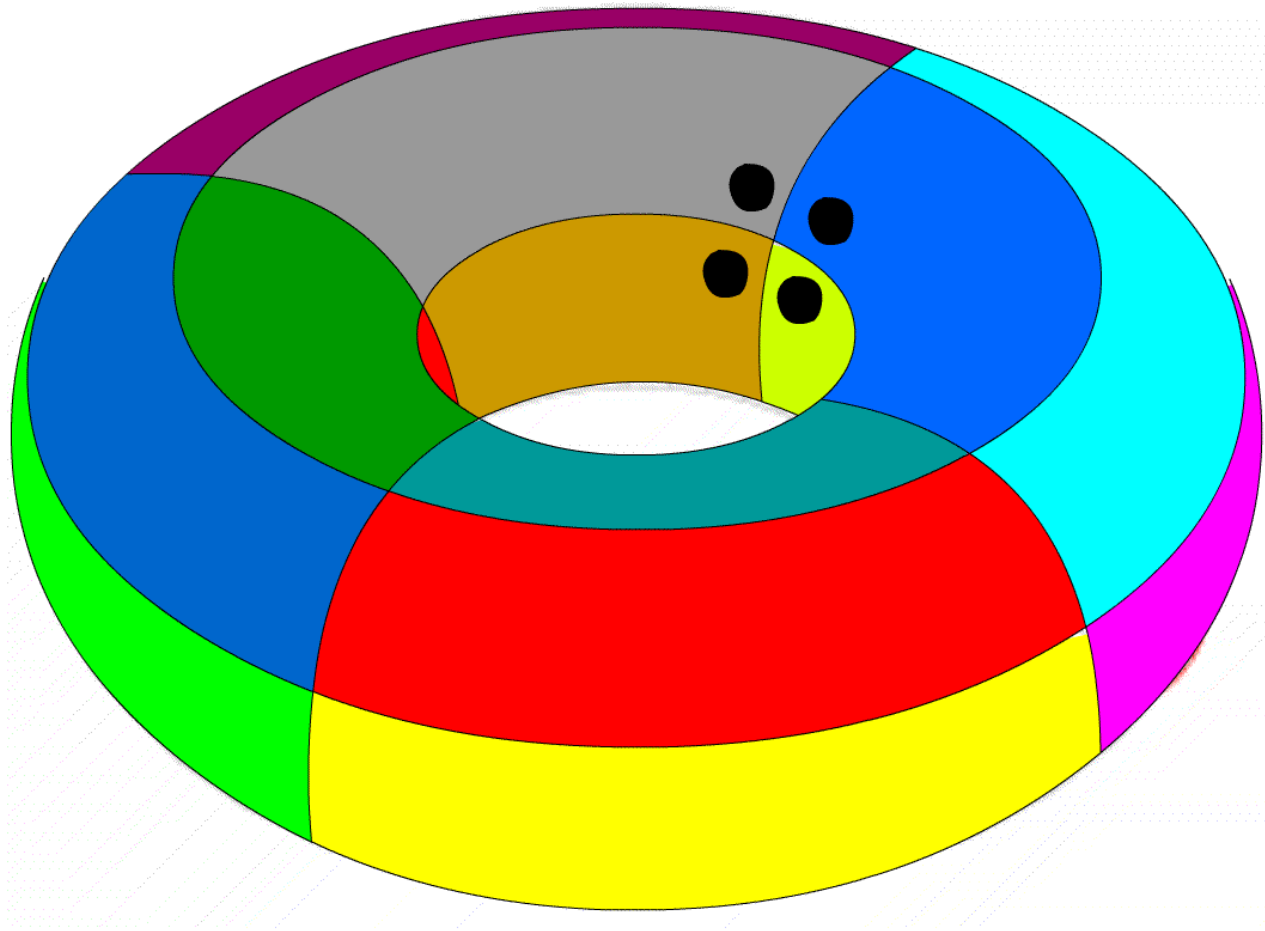
$$\begin{aligned}
 F(W,Z,Y,X) &= \prod M(3, 7, 10, 11, 15) \\
 &= (YX)'(WZ'Y)' \\
 &= (Y' + X')(W' + Z + Y')
 \end{aligned}$$



---

# MAXTERMS

---



torus

|    |    | YX                   |                      |                      |                      |
|----|----|----------------------|----------------------|----------------------|----------------------|
|    |    | 00                   | 01                   | 11                   | 10                   |
| WZ | 00 | 1<br>m <sub>0</sub>  | 0<br>m <sub>1</sub>  | 0<br>m <sub>3</sub>  | 1<br>m <sub>2</sub>  |
|    | 01 | 0<br>m <sub>4</sub>  | 0<br>m <sub>5</sub>  | 0<br>m <sub>7</sub>  | 0<br>m <sub>6</sub>  |
|    | 11 | 0<br>m <sub>12</sub> | 0<br>m <sub>13</sub> | 0<br>m <sub>15</sub> | 0<br>m <sub>14</sub> |
|    | 10 | 1<br>m <sub>8</sub>  | 0<br>m <sub>9</sub>  | 0<br>m <sub>11</sub> | 1<br>m <sub>10</sub> |

$$\begin{aligned}
 F(W,Z,Y,X) &= \sum m(0, 2, 8, 10) \\
 &= Z'X' \\
 &= \prod M(1,3-7,9,11-15) \\
 &= (X)'(Z)' \\
 &= X'Z'
 \end{aligned}$$

---

Given two unsigned numbers  $x$  and  $y$ ,  
design a logic circuit to see

$$x \geq? y$$

---

| Y2 | Y1 | X2 | X1 | F(Y2,Y1,X2,X1)=Σ m(0,1,2,3,5,6,7,10,11,15) | F(Y2,Y1,X2,X1)=Π M(4,8,9,12,13,14) |
|----|----|----|----|--------------------------------------------|------------------------------------|
| 0  | 0  | 0  | 0  | 1                                          | 1                                  |
| 0  | 0  | 0  | 1  | 1                                          | 1                                  |
| 0  | 0  | 1  | 0  | 1                                          | 1                                  |
| 0  | 0  | 1  | 1  | 1                                          | 1                                  |
| 0  | 1  | 0  | 0  | 0                                          | 0                                  |
| 0  | 1  | 0  | 1  | 1                                          | 1                                  |
| 0  | 1  | 1  | 0  | 1                                          | 1                                  |
| 0  | 1  | 1  | 1  | 1                                          | 1                                  |
| 1  | 0  | 0  | 0  | 0                                          | 0                                  |
| 1  | 0  | 0  | 1  | 0                                          | 0                                  |
| 1  | 0  | 1  | 0  | 1                                          | 1                                  |
| 1  | 0  | 1  | 1  | 1                                          | 1                                  |
| 1  | 1  | 0  | 0  | 0                                          | 0                                  |
| 1  | 1  | 0  | 1  | 0                                          | 0                                  |
| 1  | 1  | 1  | 0  | 0                                          | 0                                  |
| 1  | 1  | 1  | 1  | 1                                          | 1                                  |

|          |    | $X_2X_1$      |               |               |               |
|----------|----|---------------|---------------|---------------|---------------|
|          |    | 00            | 01            | 11            | 10            |
| $Y_2Y_1$ | 00 | 1<br>$m_0$    | 1<br>$m_1$    | 1<br>$m_3$    | 1<br>$m_2$    |
|          | 01 | 0<br>$m_4$    | 1<br>$m_5$    | 1<br>$m_7$    | 1<br>$m_6$    |
|          | 11 | 0<br>$m_{12}$ | 0<br>$m_{13}$ | 1<br>$m_{15}$ | 0<br>$m_{14}$ |
|          | 10 | 0<br>$m_8$    | 0<br>$m_9$    | 1<br>$m_{11}$ | 1<br>$m_{10}$ |

$$F(Y_2, Y_1, X_2, X_1) = \underline{\Sigma} m(0, 1, 2, 3, 5, 6, 7, 10, 11, 15)$$

$$F(Y_2, Y_1, X_2, X_1) = \underline{\Pi} M(4, 8, 9, 12, 13, 14)$$

|          |    | $X_2X_1$      |               |               |               |
|----------|----|---------------|---------------|---------------|---------------|
|          |    | 00            | 01            | 11            | 10            |
| $Y_2Y_1$ | 00 | 1<br>$m_0$    | 1<br>$m_1$    | 1<br>$m_3$    | 1<br>$m_2$    |
|          | 01 | 0<br>$m_4$    | 1<br>$m_5$    | 1<br>$m_7$    | 1<br>$m_6$    |
|          | 11 | 0<br>$m_{12}$ | 0<br>$m_{13}$ | 1<br>$m_{15}$ | 0<br>$m_{14}$ |
|          | 10 | 0<br>$m_8$    | 0<br>$m_9$    | 1<br>$m_{11}$ | 1<br>$m_{10}$ |

$$F(Y_2, Y_1, X_2, X_1) = \sum m(0, 1, 2, 3, 5, 6, 7, 10, 11, 15)$$

$$= Y_2' Y_1' +$$

|          |    | $X_2X_1$      |               |               |               |
|----------|----|---------------|---------------|---------------|---------------|
|          |    | 00            | 01            | 11            | 10            |
| $Y_2Y_1$ | 00 | 1<br>$m_0$    | 1<br>$m_1$    | 1<br>$m_3$    | 1<br>$m_2$    |
|          | 01 | 0<br>$m_4$    | 1<br>$m_5$    | 1<br>$m_7$    | 1<br>$m_6$    |
|          | 11 | 0<br>$m_{12}$ | 0<br>$m_{13}$ | 1<br>$m_{15}$ | 0<br>$m_{14}$ |
|          | 10 | 0<br>$m_8$    | 0<br>$m_9$    | 1<br>$m_{11}$ | 1<br>$m_{10}$ |

$$\begin{aligned}
 F(Y_2, Y_1, X_2, X_1) &= \sum m(0, 1, 2, 3, 5, 6, 7, 10, 11, 15) \\
 &= Y_2' Y_1' + X_2 X_1
 \end{aligned}$$

|          |    | $X_2X_1$      |               |               |               |
|----------|----|---------------|---------------|---------------|---------------|
|          |    | 00            | 01            | 11            | 10            |
| $Y_2Y_1$ | 00 | 1<br>$m_0$    | 1<br>$m_1$    | 1<br>$m_3$    | 1<br>$m_2$    |
|          | 01 | 0<br>$m_4$    | 1<br>$m_5$    | 1<br>$m_7$    | 1<br>$m_6$    |
|          | 11 | 0<br>$m_{12}$ | 0<br>$m_{13}$ | 1<br>$m_{15}$ | 0<br>$m_{14}$ |
|          | 10 | 0<br>$m_8$    | 0<br>$m_9$    | 1<br>$m_{11}$ | 1<br>$m_{10}$ |

$$\begin{aligned}
 F(Y_2, Y_1, X_2, X_1) &= \sum m(0, 1, 2, 3, 5, 6, 7, 10, 11, 15) \\
 &= Y_2'Y_1' + X_2X_1 + Y_2'X_1
 \end{aligned}$$

|          |    | $X_2X_1$      |               |               |               |
|----------|----|---------------|---------------|---------------|---------------|
|          |    | 00            | 01            | 11            | 10            |
| $Y_2Y_1$ | 00 | 1<br>$m_0$    | 1<br>$m_1$    | 1<br>$m_3$    | 1<br>$m_2$    |
|          | 01 | 0<br>$m_4$    | 1<br>$m_5$    | 1<br>$m_7$    | 1<br>$m_6$    |
|          | 11 | 0<br>$m_{12}$ | 0<br>$m_{13}$ | 1<br>$m_{15}$ | 0<br>$m_{14}$ |
|          | 10 | 0<br>$m_8$    | 0<br>$m_9$    | 1<br>$m_{11}$ | 1<br>$m_{10}$ |

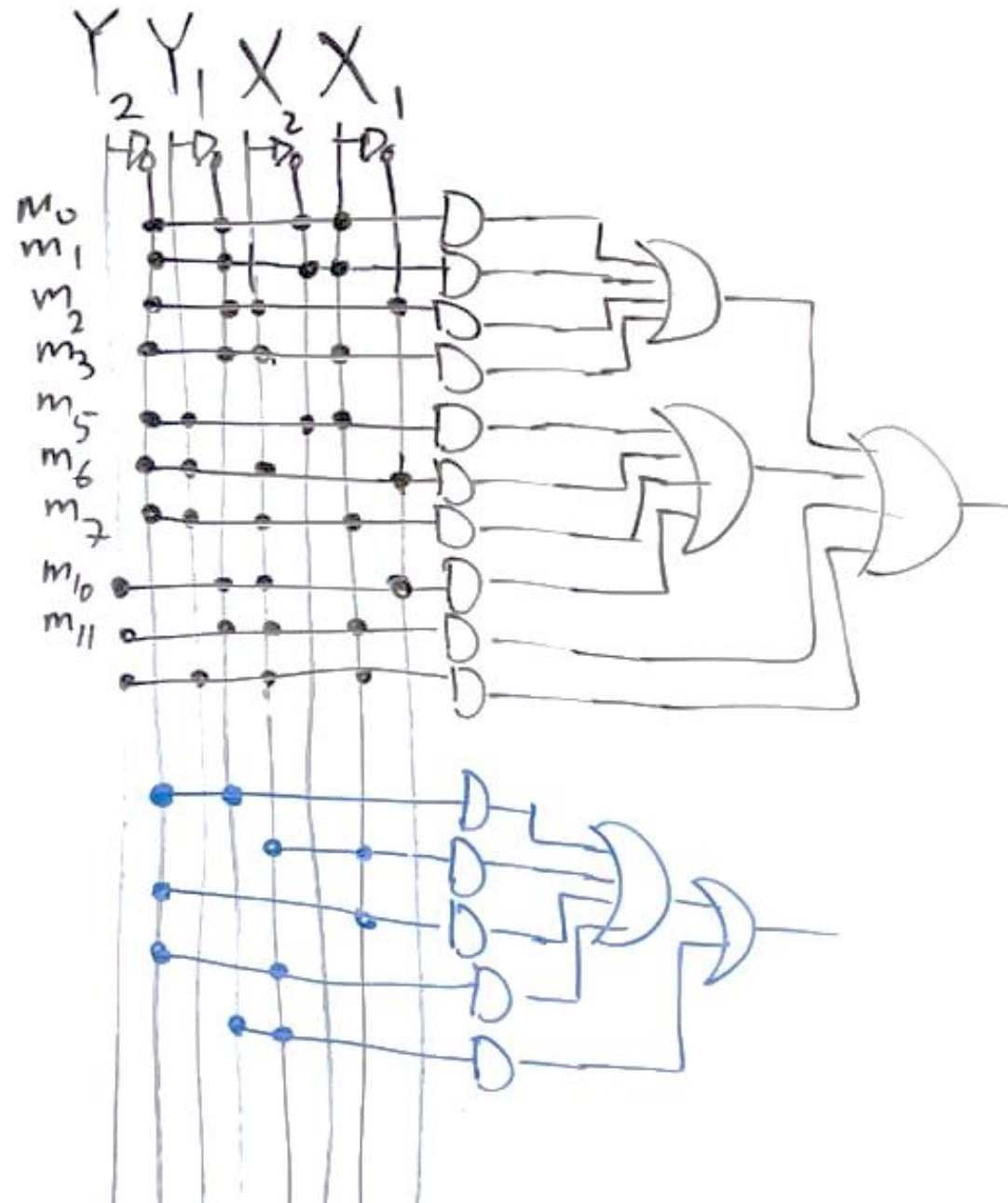
$$\begin{aligned}
 F(Y_2, Y_1, X_2, X_1) &= \sum m(0, 1, 2, 3, 5, 6, 7, 10, 11, 15) \\
 &= Y_2'Y_1' + X_2X_1 + Y_2'X_1 + Y_2'X_2
 \end{aligned}$$

|          |    | $X_2X_1$      |               |               |               |
|----------|----|---------------|---------------|---------------|---------------|
|          |    | 00            | 01            | 11            | 10            |
| $Y_2Y_1$ | 00 | 1<br>$m_0$    | 1<br>$m_1$    | 1<br>$m_3$    | 1<br>$m_2$    |
|          | 01 | 0<br>$m_4$    | 1<br>$m_5$    | 1<br>$m_7$    | 1<br>$m_6$    |
|          | 11 | 0<br>$m_{12}$ | 0<br>$m_{13}$ | 1<br>$m_{15}$ | 0<br>$m_{14}$ |
|          | 10 | 0<br>$m_8$    | 0<br>$m_9$    | 1<br>$m_{11}$ | 1<br>$m_{10}$ |

$$\begin{aligned}
 F(Y_2, Y_1, X_2, X_1) &= \sum m(0, 1, 2, 3, 5, 6, 7, 10, 11, 15) \\
 &= Y_2'Y_1' + X_2X_1 + Y_2'X_1 + Y_2'X_2 + Y_1'X_2
 \end{aligned}$$

Canonical SoP

Minimized SoP



---

# MAXTERMS

---

|          |    | $X_2X_1$      |               |               |               |
|----------|----|---------------|---------------|---------------|---------------|
|          |    | 00            | 01            | 11            | 10            |
| $Y_2Y_1$ | 00 | 1<br>$m_0$    | 1<br>$m_1$    | 1<br>$m_3$    | 1<br>$m_2$    |
|          | 01 | 0<br>$m_4$    | 1<br>$m_5$    | 1<br>$m_7$    | 1<br>$m_6$    |
|          | 11 | 0<br>$m_{12}$ | 0<br>$m_{13}$ | 1<br>$m_{15}$ | 0<br>$m_{14}$ |
|          | 10 | 0<br>$m_8$    | 0<br>$m_9$    | 1<br>$m_{11}$ | 1<br>$m_{10}$ |

$$F(Y_2, Y_1, X_2, X_1) = \prod M(4, 8, 9, 12, 13, 14)$$

|          |           | $X_2X_1$      |               |               |               |
|----------|-----------|---------------|---------------|---------------|---------------|
|          |           | <u>00</u>     | 01            | 11            | 10            |
| $Y_2Y_1$ | 00        | 1<br>$m_0$    | 1<br>$m_1$    | 1<br>$m_3$    | 1<br>$m_2$    |
|          | 01        | 0<br>$m_4$    | 1<br>$m_5$    | 1<br>$m_7$    | 1<br>$m_6$    |
|          | <u>11</u> | 0<br>$m_{12}$ | 0<br>$m_{13}$ | 1<br>$m_{15}$ | 0<br>$m_{14}$ |
|          | 10        | 0<br>$m_8$    | 0<br>$m_9$    | 1<br>$m_{11}$ | 1<br>$m_{10}$ |

$$\begin{aligned}
 F(Y_2, Y_1, X_2, X_1) &= \prod M(4, 8, 9, 12, 13, 14) \\
 &= (Y_2 X_2)'
 \end{aligned}$$

|          |    | $X_2X_1$      |               |               |               |
|----------|----|---------------|---------------|---------------|---------------|
|          |    | <u>00</u>     | 01            | 11            | 10            |
| $Y_2Y_1$ | 00 | 1<br>$m_0$    | 1<br>$m_1$    | 1<br>$m_3$    | 1<br>$m_2$    |
|          | 01 | 0<br>$m_4$    | 1<br>$m_5$    | 1<br>$m_7$    | 1<br>$m_6$    |
|          | 11 | 0<br>$m_{12}$ | 0<br>$m_{13}$ | 1<br>$m_{15}$ | 0<br>$m_{14}$ |
|          | 10 | 0<br>$m_8$    | 0<br>$m_9$    | 1<br>$m_{11}$ | 1<br>$m_{10}$ |

$$\begin{aligned}
 F(Y_2, Y_1, X_2, X_1) &= \prod M(4, 8, 9, 12, 13, 14) \\
 &= (Y_2 X_2')' (Y_1 X_2' X_1')'
 \end{aligned}$$

|           |           | $X_2 X_1$     |               |               |               |
|-----------|-----------|---------------|---------------|---------------|---------------|
|           |           | 00            | 01            | 11            | <u>10</u>     |
| $Y_2 Y_1$ | 00        | 1<br>$m_0$    | 1<br>$m_1$    | 1<br>$m_3$    | 1<br>$m_2$    |
|           | 01        | 0<br>$m_4$    | 1<br>$m_5$    | 1<br>$m_7$    | 1<br>$m_6$    |
|           | <u>11</u> | 0<br>$m_{12}$ | 0<br>$m_{13}$ | 1<br>$m_{15}$ | 0<br>$m_{14}$ |
|           | 10        | 0<br>$m_8$    | 0<br>$m_9$    | 1<br>$m_{11}$ | 1<br>$m_{10}$ |

$$\begin{aligned}
 F(Y_2, Y_1, X_2, X_1) &= \prod M(4, 8, 9, 12, 13, 14) \\
 &= (Y_2 X_2)' (Y_1 X_2' X_1)' \underline{(Y_2 Y_1 X_1)'}
 \end{aligned}$$

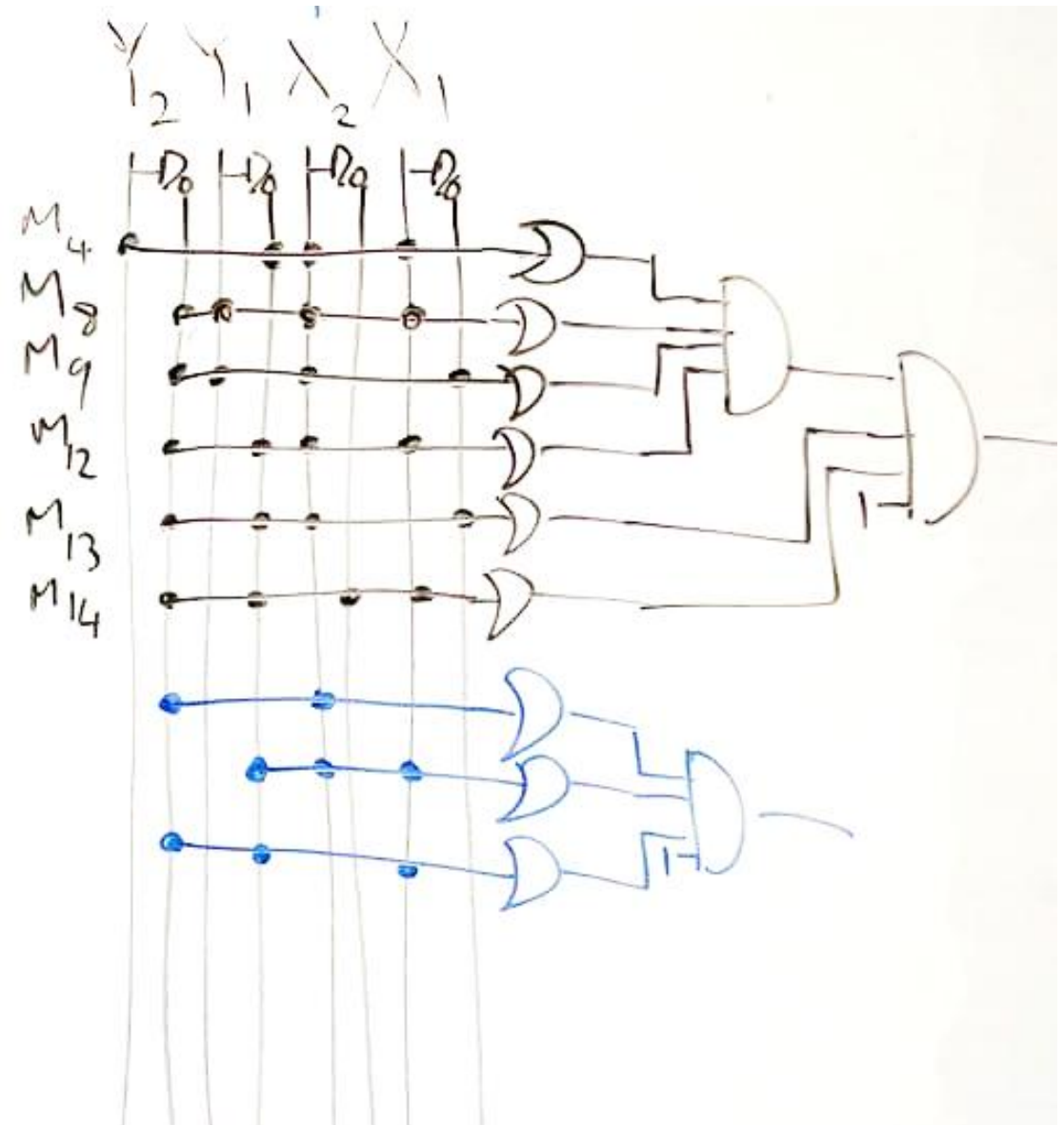
|          |    | $X_2X_1$      |               |               |               |
|----------|----|---------------|---------------|---------------|---------------|
|          |    | 00            | 01            | 11            | 10            |
| $Y_2Y_1$ | 00 | 1<br>$m_0$    | 1<br>$m_1$    | 1<br>$m_3$    | 1<br>$m_2$    |
|          | 01 | 0<br>$m_4$    | 1<br>$m_5$    | 1<br>$m_7$    | 1<br>$m_6$    |
|          | 11 | 0<br>$m_{12}$ | 0<br>$m_{13}$ | 1<br>$m_{15}$ | 0<br>$m_{14}$ |
|          | 10 | 0<br>$m_8$    | 0<br>$m_9$    | 1<br>$m_{11}$ | 1<br>$m_{10}$ |

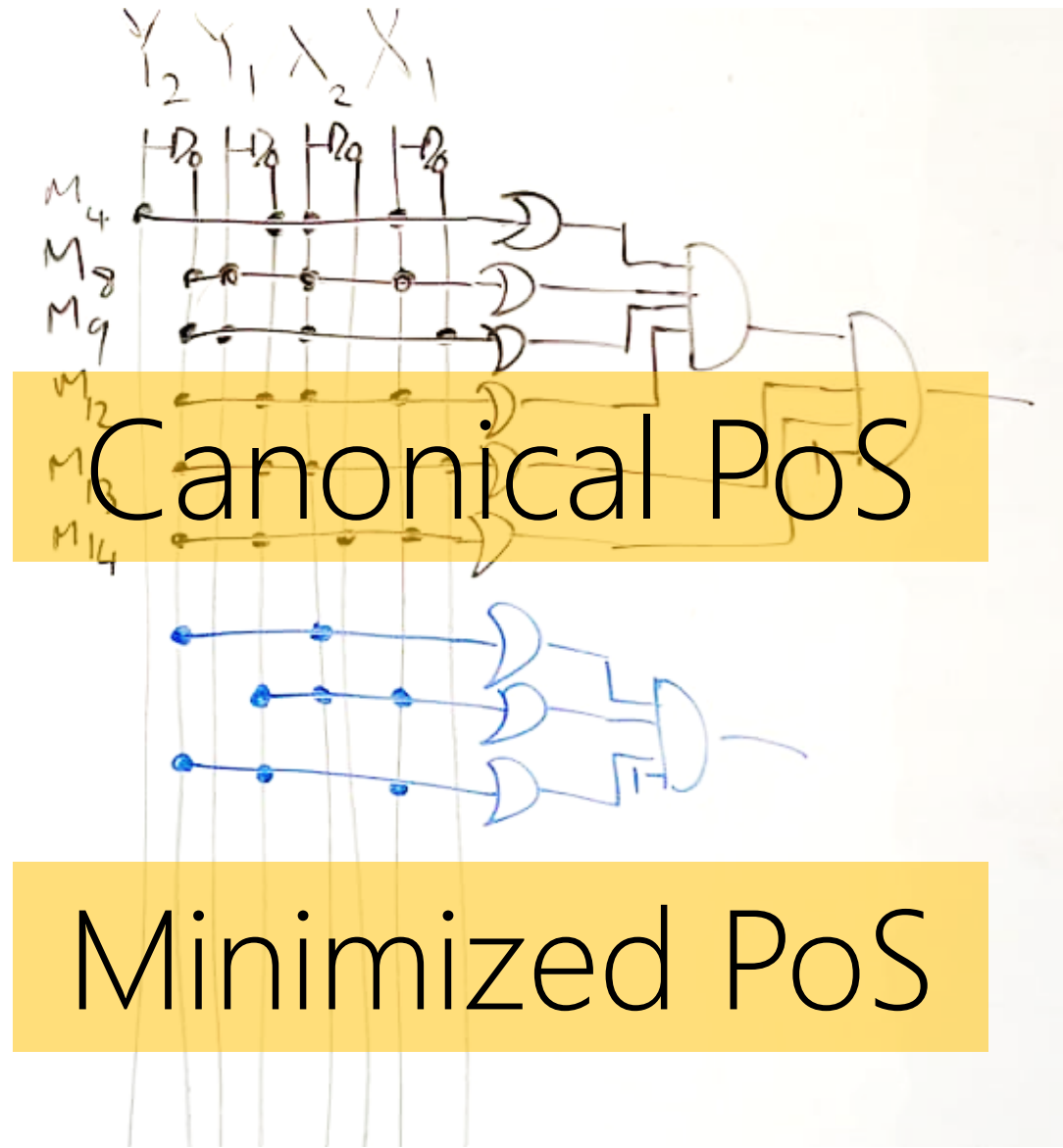
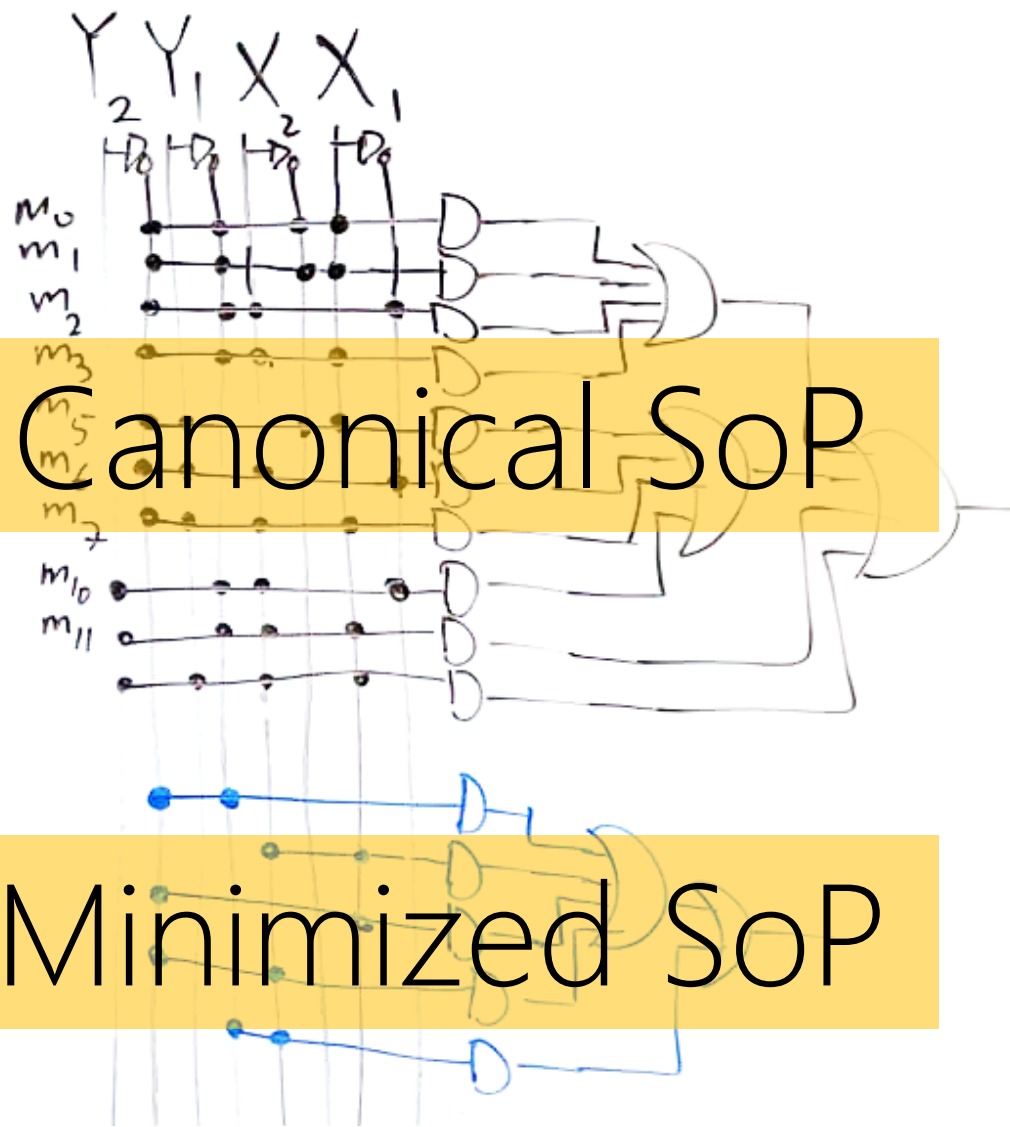
$$\begin{aligned}
 F(Y_2, Y_1, X_2, X_1) &= \prod M(4, 8, 9, 12, 13, 14) \\
 &= (Y_2X_2')' (Y_1X_2'X_1')' (Y_2Y_1X_1')' \\
 &= (Y_2' + X_2) (Y_1' + X_2 + X_1)' (Y_2' + Y_1' + X_1)
 \end{aligned}$$

(PoS)

Canonical PoS

Minimized PoS





---

# Question

---

If  $|\text{Canonical SoP}| > |\text{Canonical PoS}|$  then  $|\text{Minimized SoP}| > |\text{Minimized PoS}|$ ?

If yes, prove.

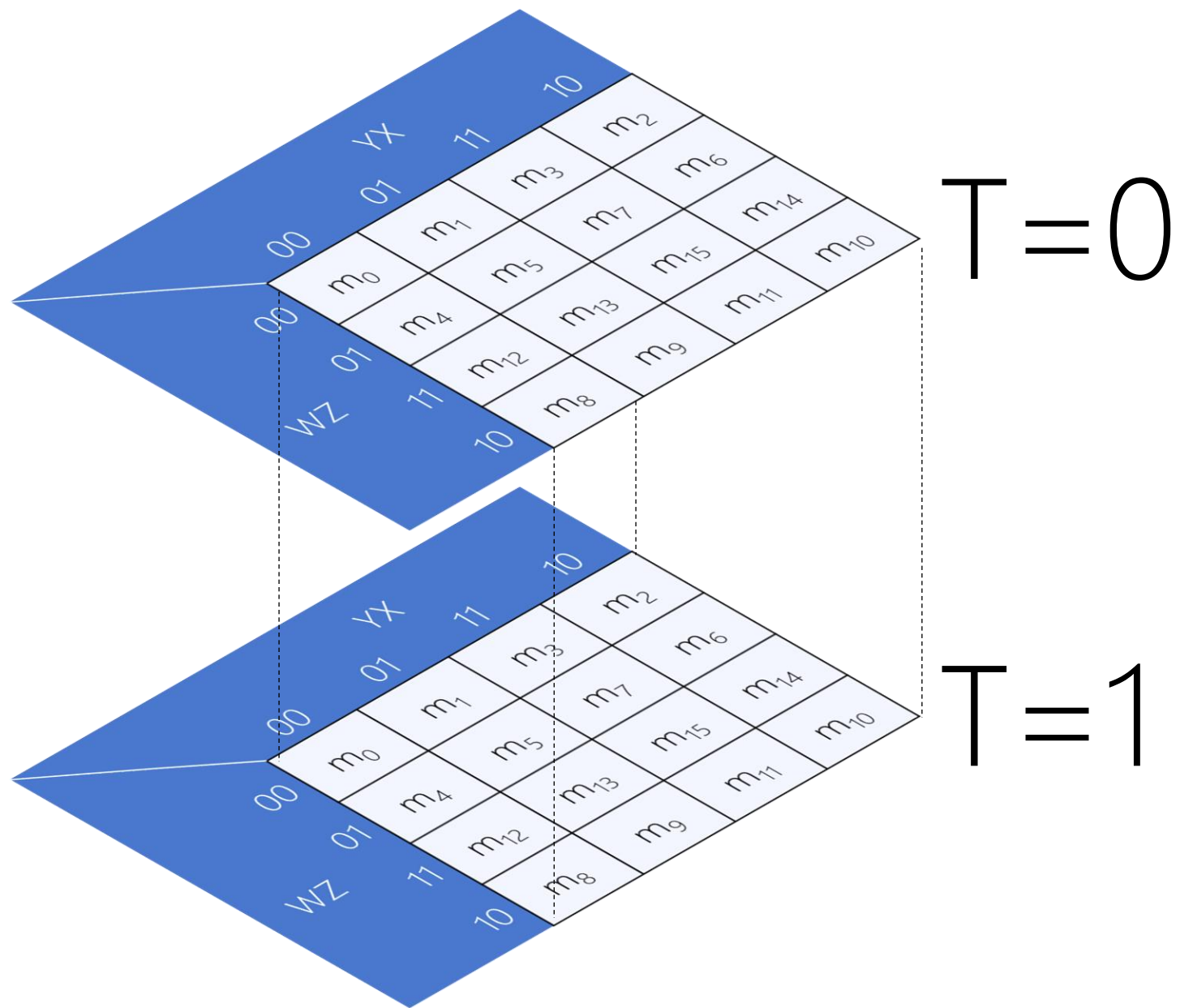
If no, bring a counter-example.

---

# 5-Variable KARNAUGH MAP

---

$$F(T, W, Z, Y, X)$$

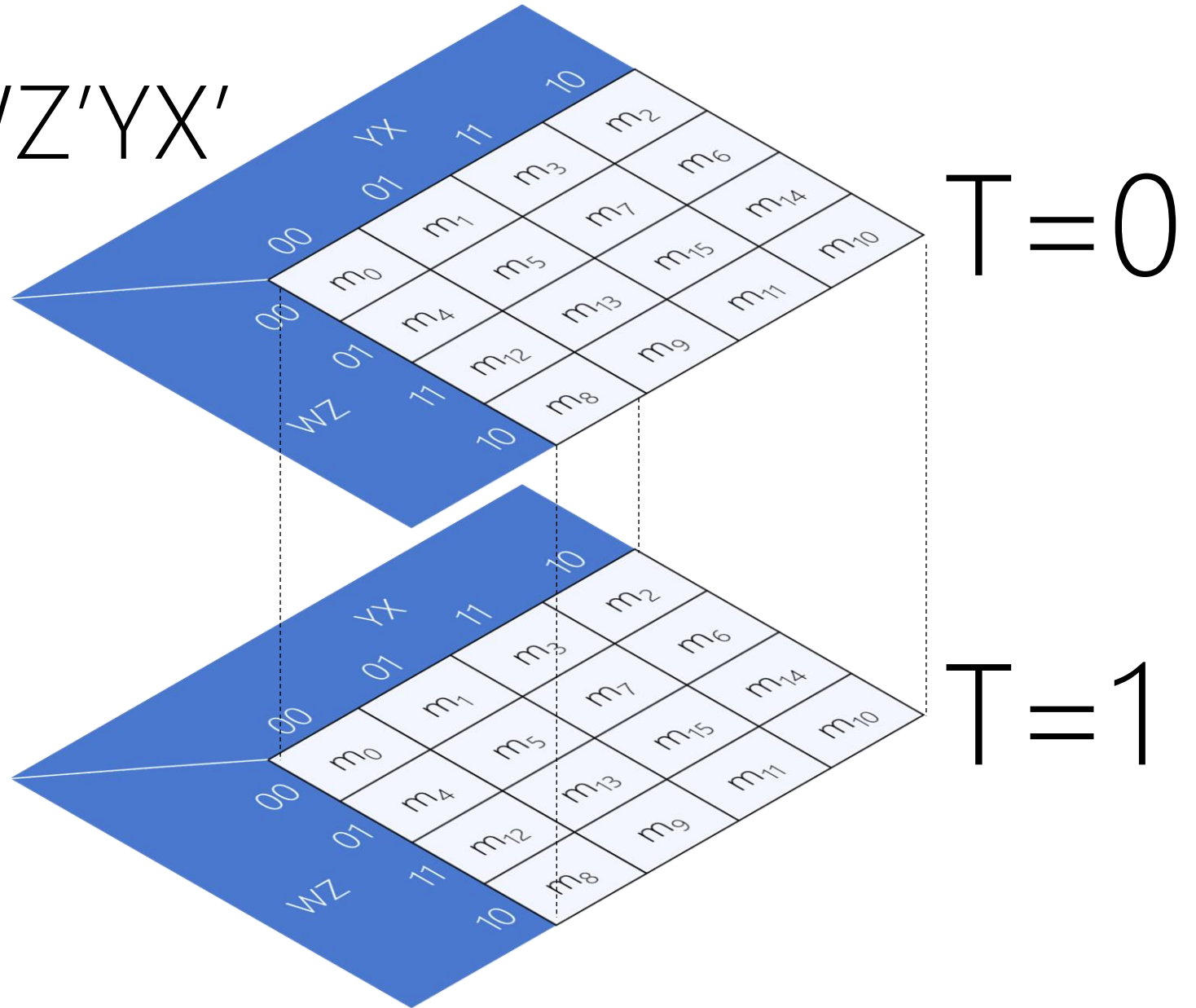


$$F(T,W,Z,Y,X)$$

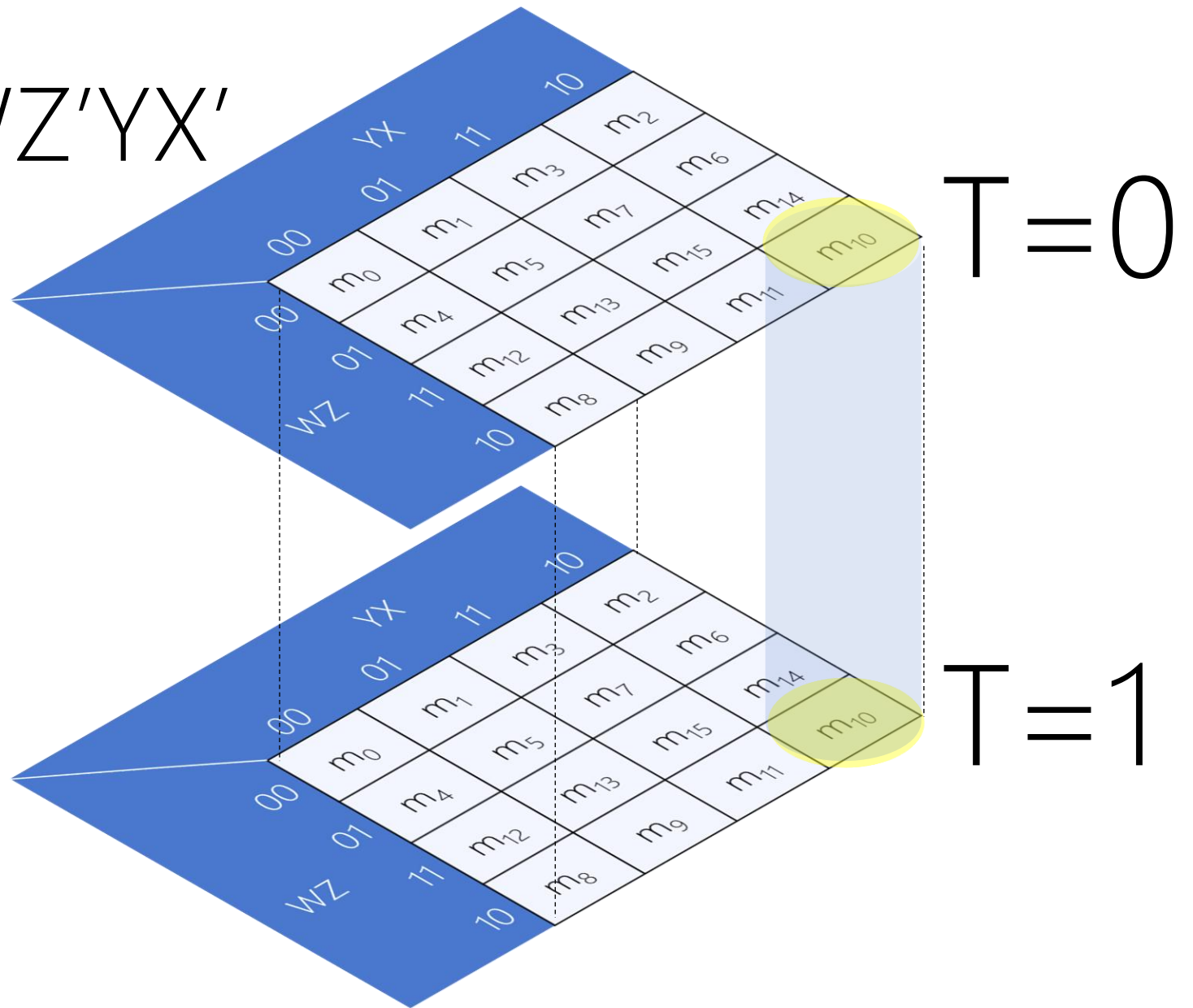
$$= T'WZ'YX' + TWZ'YX'$$

$$= m_{01010} + m_{11010}$$

$$= m_{10} + m_{26}$$



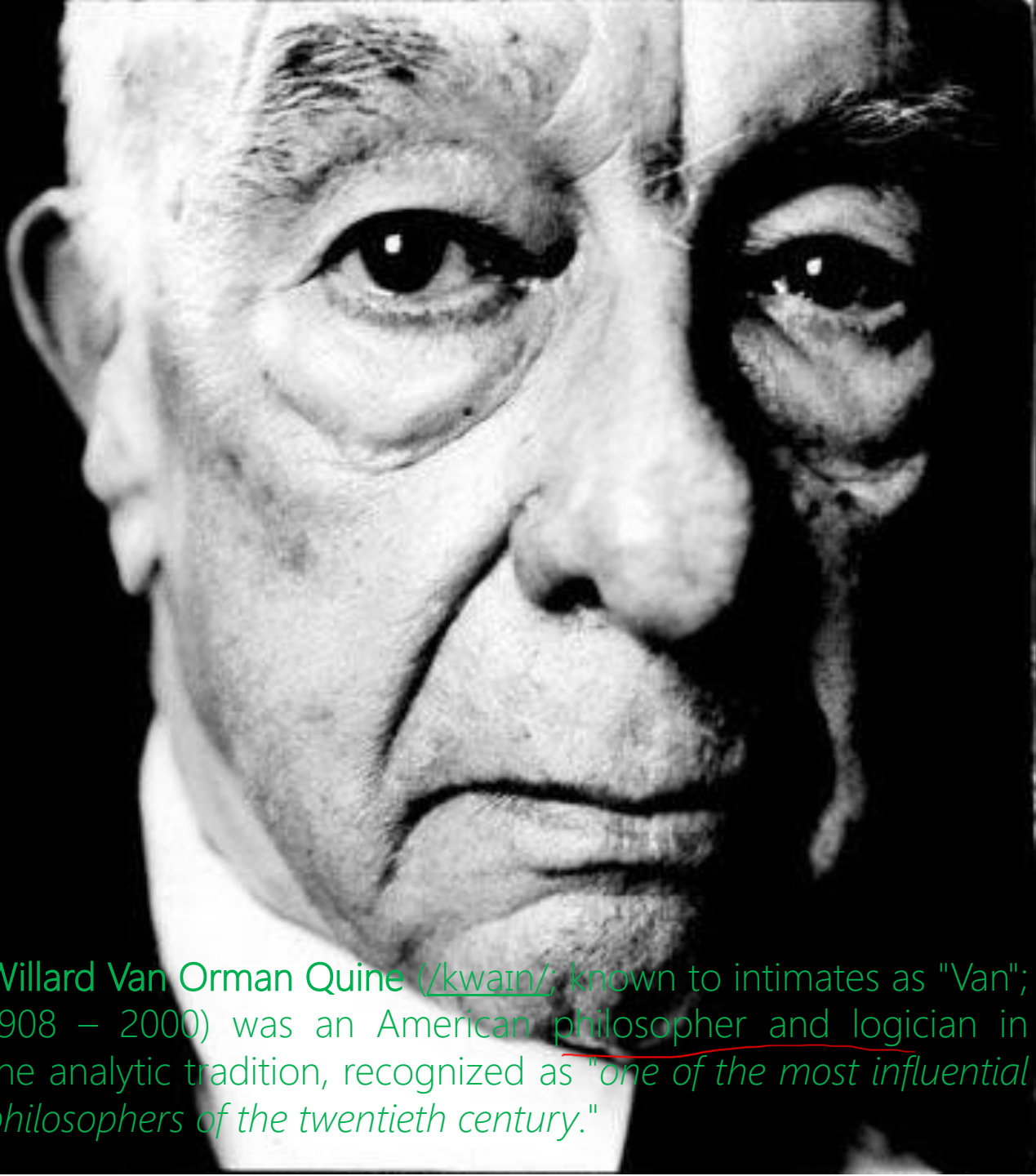
$$\begin{aligned}
 F(T,W,Z,Y,X) &= T'WZ'YX' + TWZ'YX' \\
 &= m_{01010} + m_{11010} \\
 &= m_{10} + m_{26} \\
 &= \underline{WZ'YX'}
 \end{aligned}$$



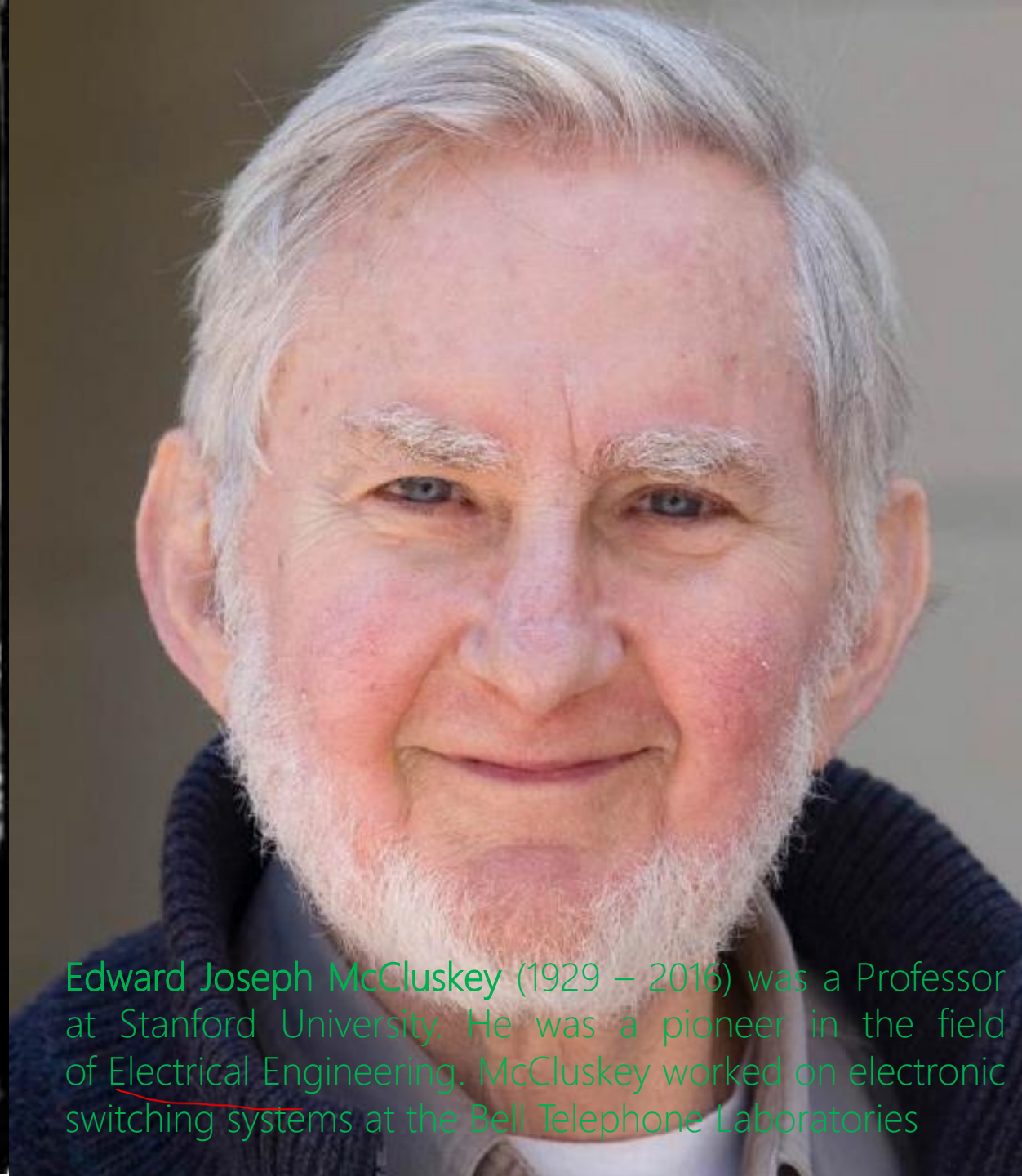
---

# n-Variable ~~KARNAUGH~~ MAP

---



Willard Van Orman Quine ([/kwain/](#); known to intimates as "Van"; 1908 – 2000) was an American philosopher and logician in the analytic tradition, recognized as *"one of the most influential philosophers of the twentieth century."*



Edward Joseph McCluskey (1929 – 2016) was a Professor at Stanford University. He was a pioneer in the field of Electrical Engineering. McCluskey worked on electronic switching systems at the Bell Telephone Laboratories

---

# Quine–McCluskey Algorithm

[https://en.wikipedia.org/wiki/Quine%E2%80%93McCluskey\\_algorithm](https://en.wikipedia.org/wiki/Quine%E2%80%93McCluskey_algorithm)

1878 ← 1937 ← 1952 ← 1956



---

# Demo

# Quine–McCluskey Algorithm

<https://www.mathematik.uni-marburg.de/~thormae/lectures/ti1/code/qmc/>

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# Don't Care Conditions

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In practice, in some applications the function is not specified for certain combinations of the variables.

| Z | Y | X | F=if input is positive(2's comp.) then 1 else 0 |
|---|---|---|-------------------------------------------------|
| 0 | 0 | 0 | ?                                               |
| 0 | 0 | 1 | ?                                               |
| 0 | 1 | 0 | ?                                               |
| 0 | 1 | 1 | ?                                               |
| 1 | 0 | 0 | ?                                               |
| 1 | 0 | 1 | ?                                               |
| 1 | 1 | 0 | ?                                               |
| 1 | 1 | 1 | ?                                               |

| Z | Y | X | F=if input is positive(2's comp.) then 1 else 0 |
|---|---|---|-------------------------------------------------|
| 0 | 0 | 0 | 0                                               |
| 0 | 0 | 1 | 1                                               |
| 0 | 1 | 0 | 1                                               |
| 0 | 1 | 1 | 1                                               |
| 1 | 0 | 0 | 0                                               |
| 1 | 0 | 1 | 0                                               |
| 1 | 1 | 0 | 0                                               |
| 1 | 1 | 1 | 0                                               |

In math, 0 is not positive nor negative!

| Z | Y | X | $F = \sum m(1,2,3) = \prod M(0,4,5,6,7)$ |
|---|---|---|------------------------------------------|
| 0 | 0 | 0 | 0                                        |
| 0 | 0 | 1 | 1                                        |
| 0 | 1 | 0 | 1                                        |
| 0 | 1 | 1 | 1                                        |
| 1 | 0 | 0 | 0                                        |
| 1 | 0 | 1 | 0                                        |
| 1 | 1 | 0 | 0                                        |
| 1 | 1 | 1 | 0                                        |

|   |   | YX         |            |            |            |
|---|---|------------|------------|------------|------------|
|   |   | 00         | 01         | 11         | 10         |
| Z | 0 | 0<br>$m_0$ | 1<br>$m_1$ | 1<br>$m_3$ | 1<br>$m_2$ |
|   | 1 | 0<br>$m_4$ | 0<br>$m_5$ | 0<br>$m_7$ | 0<br>$m_6$ |

$$\begin{aligned}
 F(Z, Y, X) &= \sum m(1, 2, 3) \\
 &= Z'X + Z'Y
 \end{aligned}$$

Boolean algebra  $\rightarrow Z'(X+Y)$

---

# MAXTERMS

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|   |   | YX         |            |            |            |
|---|---|------------|------------|------------|------------|
|   |   | 00         | 01         | 11         | 10         |
| Z | 0 | 0<br>$m_0$ | 1<br>$m_1$ | 1<br>$m_3$ | 1<br>$m_2$ |
|   | 1 | 0<br>$m_4$ | 0<br>$m_5$ | 0<br>$m_7$ | 0<br>$m_6$ |

$$\begin{aligned}
 F(Z,Y,X) &= \prod M(0,4,5,6,7) \\
 &= (Z)' (Y'X')' \\
 &= Z' (Y+X)
 \end{aligned}$$

| Z | Y | X | F=if positive(2's comp.) then 1 if negative 0 |
|---|---|---|-----------------------------------------------|
| 0 | 0 | 0 | 0                                             |
| 0 | 0 | 1 | 1                                             |
| 0 | 1 | 0 | 1                                             |
| 0 | 1 | 1 | 1                                             |
| 1 | 0 | 0 | 0                                             |
| 1 | 0 | 1 | 0                                             |
| 1 | 1 | 0 | 0                                             |
| 1 | 1 | 1 | 0                                             |

In math, 0 is not positive nor negative!

| Z | Y | X | F=if positive(2's comp.) then 1 if negative 0 |
|---|---|---|-----------------------------------------------|
| 0 | 0 | 0 | 0                                             |
| 0 | 0 | 1 | 1                                             |
| 0 | 1 | 0 | 1                                             |
| 0 | 1 | 1 | 1                                             |
| 1 | 0 | 0 | 0                                             |
| 1 | 0 | 1 | 0                                             |
| 1 | 1 | 0 | 0                                             |
| 1 | 1 | 1 | 0                                             |

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# Don't Care Conditions

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Functions that have unspecified outputs for some input combinations are called *incompletely specified functions*.

Don't-care conditions can be used on a map to provide further simplification of the Boolean expression.

|   |   | YX                                                                                           |            |            |            |
|---|---|----------------------------------------------------------------------------------------------|------------|------------|------------|
|   |   | 00                                                                                           | 01         | 11         | 10         |
| Z | 0 | <br>$m_0$ | 1<br>$m_1$ | 1<br>$m_3$ | 1<br>$m_2$ |
|   | 1 | 0<br>$m_4$                                                                                   | 0<br>$m_5$ | 0<br>$m_7$ | 0<br>$m_6$ |

$$\begin{aligned}
 F(Z, Y, X) &= \sum m(1, 2, 3) + \sum d(0) \\
 &= Z'X + Z'Y
 \end{aligned}$$

|   |   | YX         |            |            |            |
|---|---|------------|------------|------------|------------|
|   |   | 00         | 01         | 11         | 10         |
| Z | 0 | 1<br>$m_0$ | 1<br>$m_1$ | 1<br>$m_3$ | 1<br>$m_2$ |
|   | 1 | 0<br>$m_4$ | 0<br>$m_5$ | 0<br>$m_7$ | 0<br>$m_6$ |

$$\begin{aligned}
 F(Z,Y,X) &= \sum m(1, 2, 3) + \sum m(0) \\
 &= Z'
 \end{aligned}$$

In this case, the don't care condition help to more simplification

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# MAXTERMS

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|   |   | YX         |            |            |            |
|---|---|------------|------------|------------|------------|
|   |   | 00         | 01         | 11         | 10         |
| Z | 0 | 1<br>$m_0$ | 1<br>$m_1$ | 1<br>$m_3$ | 1<br>$m_2$ |
|   | 1 | 0<br>$m_4$ | 0<br>$m_5$ | 0<br>$m_7$ | 0<br>$m_6$ |

$$\begin{aligned}
 F(Z,Y,X) &= \prod M(0,4,5,6,7) + \sum M(0) \\
 &= (Z)'(Y'X')' \\
 &= Z' (Y+X)
 \end{aligned}$$

In this case, the don't care condition does NOT help to more simplification

|   |   | YX                                                                                           |            |            |            |
|---|---|----------------------------------------------------------------------------------------------|------------|------------|------------|
|   |   | 00                                                                                           | 01         | 11         | 10         |
| Z | 0 | <br>$m_0$ | 1<br>$m_1$ | 1<br>$m_3$ | 1<br>$m_2$ |
|   | 1 | 0<br>$m_4$                                                                                   | 0<br>$m_5$ | 0<br>$m_7$ | 0<br>$m_6$ |

$$\begin{aligned}
 F(Z,Y,X) &= \prod M(4,5,6,7) + \sum D(0) \\
 &= (Z)' \\
 &= Z'
 \end{aligned}$$

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# Don't Care Conditions

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To distinguish the don't-care condition from 1's and 0's, an **X** is used.

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|   |   | YX                |            |            |            |
|---|---|-------------------|------------|------------|------------|
|   |   | 00                | 01         | 11         | 10         |
| Z | 0 | <b>X</b><br>$m_0$ | 1<br>$m_1$ | 1<br>$m_3$ | 1<br>$m_2$ |
|   | 1 | 0<br>$m_4$        | 0<br>$m_5$ | 0<br>$m_7$ | 0<br>$m_6$ |

$$F(Z,Y,X) = \sum m(1, 2, 3) + \sum d(0)$$

$$F(Z,Y,X) = \prod M(4,5,6,7) + \prod D(0)$$